

Let P be a poset. For $x, y \in P$, we say x **covers** y if $x > y$ and there is no z such that $x > z > y$.

A **grading** on P is a function $r: P \rightarrow \mathbb{Z}$ such that

(a) $r(x) > r(y)$ if $x > y$.

(b) $r(x) = r(y) + 1$ if x covers y .

Prop: If P is a polyhedron, then $F \mapsto \dim F$ is a grading on $F(P)$.

Proof: Suppose $F, G \in F(P)$, G covers F .

$$[F, G] = \{ H \in \mathcal{F}(P) : F \leq H \leq G \}$$

has two elements F, G .

$$[F, G] \cong \mathcal{F}(D(G, F))$$

$D(G, F)$ is a polyhedral cone. In particular, it is closed.

$$D(G, F) = C \oplus L, \quad L = \text{lin} D(G, F), \quad \text{lin} C = \{0\}$$

C is a closed cone, so C is the conical hull of its extreme rays.

$$F(C) \cong F(C \oplus L) \cong [F, G] \leftarrow \text{two elements}$$

So C itself is a ray.

$$F(C) = \begin{matrix} C \\ 1 \\ 0 \end{matrix} \leftarrow \text{diff dim}$$

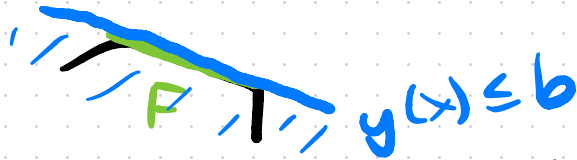
$$\dim G = \dim D(G, F)$$

$$\dim F = \dim D(F, F)$$

$$F(C \oplus L) = \begin{matrix} C \oplus L = D(G, F) \\ 1 \leftarrow \text{diff dim} \\ L = D(F, F) \end{matrix}$$

$$\text{So } \dim G - \dim F = 1$$

Given a polyhedron of dimension d , the $(d-1)$ -dimensional faces of P are called **facets** of P . If F is a facet, an inequality $y(x) \leq b$ such that $\arg\max_P y = F$ and $y(F) = b$ is called a **facet-defining inequality** of P for F .



Thm: Let P be a polyhedron. The following are true.

(a) For each facet F of P , let $y_F(x) \leq b_F$ be a facet-defining inequality. Then

$$P = \{x \in V : y_F(x) \leq b_F \quad \forall F\} \cap \text{aff } P$$

(b) Assume

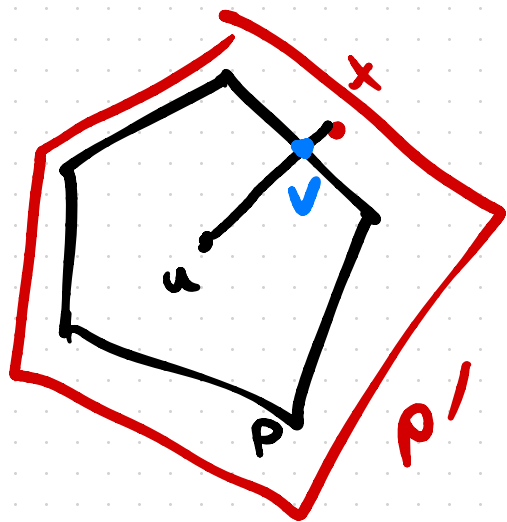
$$P = \{x \in V : y_i(x) \leq b_i \quad \forall i \in I\}$$

where I is finite. Then for every facet F of P , there is some $i \in I$ such that $y_i(x) \leq b_i$ is a facet-defining inequality for F .

Proof:

(a) Let $P' = \{x \in V : y_F(x) \leq b_F \quad \forall F\} \cap \text{aff } P$

Clearly $P \subset P'$. Suppose $x \in P' \setminus P$.



Let $u \in \text{rint} P$. The segment $[u, x]$ intersects the boundary of P at some v .

Claim: v is in a facet G of P .

Proof: v is in the relative interior of some face F of P , and this face is not P . Because $F(P)$ is graded by dimension, F is contained in some facet G of P .

By assumption, $y_G(x) \leq b_G$ is facet-defining for b .

$y_G(u) < b_G$ since $u \in P \setminus G$.

$$y_G(v) = b_G$$

$\Rightarrow y_G(x) > b_G$ contradicting def of P .

(b) Assume

$$P = \{x \in V : y_i(x) \leq b_i \quad \forall i \in I\}$$

where I is finite.

Let $\bar{F}$ be a facet of P , we have

$$F = P \cap \underbrace{\{x \in U : y_j(x) = b_j \quad \forall j \in J\}}_{:= A}$$

for some $J \subset I$. Since $\dim F = \dim P - 1$, A must intersect $\text{aff} P$ in a hyperplane of $\text{aff} P$.

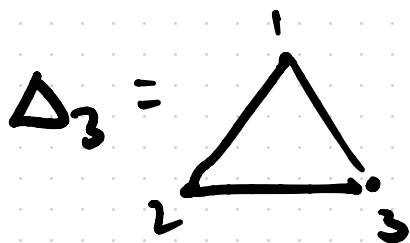
Thus we can find a single $j \in J$ such that

$$\text{aff} P \cap A = \text{aff} P \cap \{y_j(x) = b_j\}$$

Then $F = P \cap \{y_j(x) = b_j\}$, so $y_j(x) \leq b_j$ is a facet-defining inequality for $\overline{F}$.

Simplex:

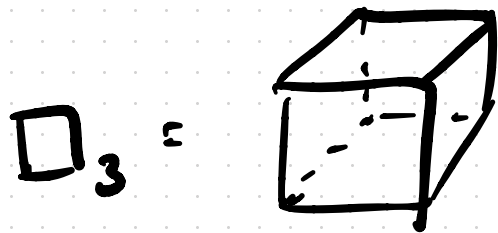
The convex hull of a set of n affinely independent points.



$$F(\Delta_n) \cong 2^{[n]} / \{\emptyset\}$$

Cube

Convex hull of the 2^n points $(\pm 1, \pm 1, \dots, \pm 1)$



$$\square_n = \{x \in \mathbb{R}^n : -1 \leq x_i \leq 1 \quad \forall i\} \quad 2^n \text{ total faces}$$

Every face is obtained by setting $x_j = 1$, $x_k = -1$
for all $j \in J \subset I$, $k \in K \subset I$, $J \cap K = \emptyset$.

Cross-polytope

Convex hull of $\pm e_i$, e_i standard basis vectors of $\mathbb{R}^n$.



Facet description:

$$\{x \in \mathbb{R}^n : \pm x_1 \pm x_2 \pm \dots \pm x_n \leq 1\}$$

Faces are convex hulls of $\{e_j\}_{j \in J} \cup \{-e_k\}_{k \in K, J \cap K = \emptyset}$, $J, K \subset [n]$

There are 3^n trees

$JUK \neq \emptyset$
and the whole Δ_n